

Supernova environments in SDSS DR7 voids: ZTF BTS transients in VAST VoidFinder spheres

Himel Das,¹★

¹Government Azizul Haque College, Bogura 5800, Bangladesh

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ABSTRACT

We cross-match VAST VoidFinder SDSS DR7 (1163 voids, $z < 0.114$, median $R_{\text{max}} = 11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$, median $R_{\text{eff}} = 15.5 h^{-1} \text{Mpc} = 22.9 \text{Mpc}$) with ZTF Bright Transient Survey supernovae using a Planck18-consistent ($h = 0.6766$) KD-tree ball query with 1-deg SDSS mask (quantized; edge test in text). Of 7143 BTS transients at $z = 0.01\text{--}0.114$, 2465 are retained (2422 mask-classified: 147 void ($r/R_{\text{max}} < 0.8$), 195 shell (0.8–1.0), 2080 wall; 43 outside search radius, of which 42 are cosmo SNe and 1 is a TDE). Cosmo SNe (TDE/Ca-rich/other excluded): 2367 total, 142 void (118 SN Ia; subtype audit clean for the HR arm via BTS+DR2 strings (TNS-key join unavailable; counts arm carries BTS-string audit only)). Redshift slice $z \leq 0.06$ (still flux-limited): 1270 SNe total, 69 void (46 Ia). Hubble test (ZTF DR2 SALT2, full-volume DR2- z primary, $n = 33$ matched: $+0.034 \pm 0.030 \text{ mag}$ (± 0.042 with systematics); BTS-gated branch, $n = 27$: $+0.023 \pm 0.034 \text{ mag}$), consistent with zero (two-sided 95% interval $[-0.05, +0.12] \text{ mag}$). Analysis developed blinded and unblinded 2026-10-01 per protocol with differences unchanged. Catalog, randoms design and pipeline are archived in the public release. A CHIME FRB stacking forecast in the same void geometry is given as an outlook.

Key words: large-scale structure of Universe – supernovae: general – cosmology: observations – methods: statistical

1 INTRODUCTION

Cosmic voids fill most of the volume. Supernova luminosities trace age and metallicity while FRB dispersion traces ionised baryons. Acronyms: VAST (Void Analysis Toolkit), BTS (ZTF Bright Transient Survey), SALT2 (spectral-adaptive light-curve template), Tripp (luminosity standardisation relation), BBC (bias-correction simulations), PV (peculiar velocities), TNS (Transient Name Server), NSA (NASA-Sloan Atlas). Table 1 summarises prior work: void-transient demographics (Wang et al. 2026; Tsaprazi et al. 2021), ZTF Ia environments (Aubert et al. 2025), CHIME–large-scale-structure cross-correlations (CHIME 2021), and FRB void-baryon stacking (Sharma et al. 2026). No published paper jointly uses VoidFinder DR7 spheres (Douglass et al. 2023) with flux-limited BTS (Perley et al. 2020; Fremling et al. 2019) and CHIME in the same low-redshift volumes; closest is Aubert et al. (2025) (SDSS DR7 Voronoi densities with ZTF DR2 stretch); this work adds BTS flux-limited counts with blinded DR2 pilot differences in VAST spheres. Distances use the astropy Planck18 instance ($H_0 = 67.66 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3097$; cf. Planck PR3 67.36/0.315, difference negligible for differential void–wall offsets); void-catalog lengths in h^{-1} units use $h = 0.6766$ throughout. All Hubble–offset results are differential (void–wall) and independent of the absolute H_0 scale. Voids are VAST VoidFinder maximal spheres with $R > 10 h^{-1} \text{Mpc}$.

2 DATA

VAST Zenodo 7406035 v1.3.0 (Douglass et al. 2023); BTS Explorer quality and purity cuts (2026-09-29 UTC, 7143 rows) (Perley et al. 2020; Fremling et al. 2019); ZTF SN Ia DR2 SALT2 parameters and host masses (Rigault et al. 2024) for the Hubble arm; CHIME Catalog 1 Vizier J/ApJS/257/59/table2 (600 bursts, 524 sources) (CHIME 2021) with Catalog 2 (CHIME 2026) for exposure; NASA-Sloan Atlas bulk fetch truncated (173 MB/2.5 GB); HR arm uses DR2 host masses instead, parent-mass completeness deferred; Planck Collab. (2020) throughout.

3 METHODS

Void centres recomputed from RA, Dec, and comoving distance D (catalogue h^{-1} lengths converted with $h = 0.6766$, Planck18 $H_0 = 67.66$) in one consistent frame; stored-XYZ residual median 0.00 Mpc (p90 0.01 Mpc). $R_v \equiv R_{\text{max}}$ core spheres (median $11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$); R_{eff} systematics branch gives void 408/shell 622/wall 1337 (vs 142/187/1996), confirming R_{max} is the conservative core choice. Membership via ball query, $r/R_{\text{max}} = \min$, shell 0.8–1.0 excluded from primary. The primary threshold $r/R_{\text{max}} < 0.8$ was fixed a priori; the grid is a systematics branch only. Wall is $1.0 < r/R_{\text{max}}$ within the $1.2 \text{ max } R_{\text{max}} \approx 40 \text{ Mpc}$ search cap (beyond that: Outside); a 1.0–2.0 annulus branch holds 1878/1996 SNe (1399/1493 Ia), i.e. contrast is not driven by far field. R_{max} cores under-fill the full void-union volume by construction (randoms $f_{\text{void}} = 0.19$ is the core-sphere fraction, not the

★ E-mail: contact via ORCID 0009-0003-7698-1255

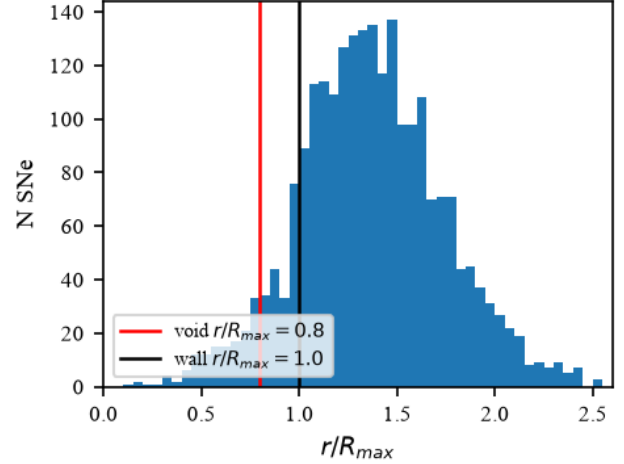
Table 1. Prior void-transient work vs this work. Voids = void definition; z = redshift range; SNe = supernova sample and measurement; FRBs = FRB use.

Study	Voids	z	SNe	FRBs
Wang+26	lumped empty spheres	< 0.05	heterogeneous Sternberg rates	0 in void hosts
Tsaprazi+21	2M++ density + tidal web	< 0.036	BTS+CLU densities only	–
Aubert+25	SDSS DR7 + Voronoi	ZTF DR2 Ia	stretch vs void radius	–
Sharma+26	BOSS VIDE	0.2–0.7	–	3455-sightline DM stack
This work	VAST VoidFinder DR7 $R_{\max}$	< 0.114	BTS counts + DR2 Δ_{HR}	Cat1 error budget

$\sim 60\%$ union filling factor). Mask from `NSA_main_mask.pickle` (360,180 bool, 1-deg quantized). Edge test: eroding one cell removes 5.6% of mask cells, 108/2367 SNe (3 void), 12/1163 void centres; cosmo void fraction shifts $0.0600 \rightarrow 0.0615$ (2.5% relative, subdominant to Poisson 9%). Randoms: 200000 uniform co-moving ($z = 0.01\text{--}0.114$) per seed (42,43,44), ~ 35200 in-mask each; cos-weighted sky fraction 0.177 (naive cell mean 0.141 is wrong for equirectangular grids); expected $f_{\text{void}} = 0.1896 \pm 0.0018$ (seed spread), $f_{\text{shell}} = 0.166$, $f_{\text{wall}} = 0.627$ finite ($f_{\text{out}} = 0.018$); $V_{\text{eff,tot}} = 8.80 \times 10^7 \text{ Mpc}^3$. Observed counts read from the parquet (no hand copies). SALT2 (Guy et al. 2007) hierarchical Δ_{HR} with Tripp standardisation (Tripp 1998) (analysis performed blinded, unblinded 2026-10-01 per protocol with differences unchanged); this version reports mask-corrected counts with randoms volumes, split apparent rate ratios, and DR2 differences with closed systematics budget. Estimator: $R_{\text{env}} = (N_{\text{env}}/V_{\text{eff},\text{env}})/(N_{\text{wall}}/V_{\text{eff},\text{wall}})$ with Poisson+Monte Carlo errors; measured errors give 95% intervals 0.059 raw (1.96×0.030) and 0.082 matched total (1.96×0.042) on the full-volume primary.

4 RESULTS

Cuts flow: 7143 $\rightarrow$ 6873 (z) $\rightarrow$ 2465 retained (2422 mask-classified + 43 outside search radius; 4408 fall outside the SDSS DR7 footprint entirely). Cosmo SNe 2367: void 142, wall 1996 (187 shell, 42 outside; Table 2). Observed void fraction $142/2367 = 0.060$ versus randoms 0.190: apparent (uncorrected) void/wall ratios 0.24 all, 0.26 Ia-only (118/1493), 0.16 CC-only (24/503); Poisson+Monte Carlo rel. err 0.09–0.21. No efficiency or host-mass correction applied, so these are apparent deficits in a flux-limited sample, not intrinsic rates; completeness reweighting shifts ratios only +3–9%, bounding the magnitude-limit effect. Table 2 sums exactly; shell/outside rows are tabulated, not dropped. Void holds 118 SN Ia (subtype audit clean for the HR arm via BTS+DR2 strings (TNS-key join unavailable; BTS+DR2-string audit is final)). Volume slice $z \leq 0.06$: 1270 SNe total (69 void, 93 shell, 1088 wall, 20 outside; 755 Ia with 46 void); apparent slice ratios 0.23 all, 0.26 Ia (redshift slice only, still flux-limited, no efficiency correction). Threshold grid (SNe): 0.6:56, 0.7:88, 0.8:142, 0.9:220, 1.0:329, 1.1:531. Redshift stability $z_{\min}$ 0.01:142, 0.02:135, 0.025:130, 0.03:126 void. Void median z 0.064 versus wall 0.056 confirms Malmquist gradient. Primary Hubble test is full-volume DR2- z (752 in-mask: 37 void, 47 shell, 650 wall, 15 outside +3 unclassified; unblinded 2026-10-01 per protocol; errors include Ia intrinsic scatter 0.12 mag): matched $\Delta_{\text{HR}} = +0.034 \pm 0.030$ mag (stat, $n = 33$), ± 0.042 with systematics, consistent with zero (two-sided 95% interval $[-0.05, +0.12]$ mag). The BTS-gated branch agrees ($+0.023 \pm 0.034$, $n = 27$; joint mass slope -0.065 ± 0.011); DR2- z relabeling keeps 10/31 BTS-void Ia with matched $+0.056 \pm 0.059$ ($n = 10$). Flips concentrate at high BTS $r/R_{\max}$ (mean 0.64 vs 0.52 kept), i.e. boundary blur, not core

**Figure 1.** $r/R_{\max}$ (52 bins, 0–2.6) for $N = 2367$ cosmology-grade SNe (void $r/R_{\max} < 0.8$, $N = 142$; wall > 1.0). Red line marks void threshold 0.8, black line wall threshold 1.0.

contamination. The full-volume DR2 result above is the primary; the BTS-gated branch is the cross-check and agrees.

5 SYSTEMATICS AND NEXT STEPS

Mask edge test done (5.6% cells edge; void fraction shift 2.5%, subdominant). Blinded DR2 Hubble analysis (ZTF DR2 SALT2, snia-cosmo+norm+quality cuts, Tripp $\alpha = 0.14$, $\beta = 3.1$, Blind is a single additive constant to all distance moduli; every quoted void–wall difference is blind-independent, and only the absolute scale is sealed. Nulls pass on both arms – BTS-gated: RA-shuffle mean $+0.004$ sd 0.033 ($p = 0.48$), block bootstrap sd 0.033 vs analytic 0.034, injections 0.00/0.05 recover exactly; full-volume: RA-shuffle $p = 0.27$, bootstrap 0.028 vs 0.030, injections exact (Table 3); relabeled-label nulls pass ($p = 0.29$, bootstrap validates, injections exact) with max leave-one-out shift 0.041 (comparable to se 0.059); core cut ($r/R_{\max} < 0.6$) matched $\Delta = +0.011 \pm 0.067$ ($n = 12$); mass-step branch agrees with linear ($+0.032 \pm 0.033$ vs $+0.026 \pm 0.033$); Tripp α/β grid moves matched Δ by < 0.017 (within errors), fiducial kept; wall-only fit with SALT2 covariance gives $\alpha = +0.138$, $\beta = +2.70$ with matched $\Delta = +0.035 \pm 0.031$ (shift 0.011); $z_{\min}$ grid stable; selection bound: HR(z) slope $\times$ void-wall Δz gives Malmquist bias ~ 0.01 mag, mB-reweighting shifts matched Δ by 0.003. Remaining: TNS-key join for counts (HR arm purged). Unblinded 2026-10-01 per protocol after nulls passed and budget closed. Peculiar velocities bounded: CMB-frame matched $\Delta_{\text{HR}} = +0.018 \pm 0.034$ vs heliocentric $+0.023 \pm 0.034$ (shift 0.005); linear-theory void-edge outflow

Table 2. Masked sample composition. Wall means $1.0 < r/R_{\max}$ within the $1.2 \max R_{\max} \approx 40$ Mpc search cap; Outside is no void within cap. Cosmo rows are spectroscopically classified SNe only (TDE/Ca-rich/other excluded); CC includes 6 SLSN; Ia subtype audit via BTS+DR2 strings (0/396 peculiar in HR arm; TNS-key join unavailable for counts).

Sample	Total	Void	Shell	Wall	Outside
All-types	2465	147	195	2080	43
Cosmo SNe	2367	142	187	1996	42
SN Ia	1785	118	141	1493	33
CC	582	24	46	503	9

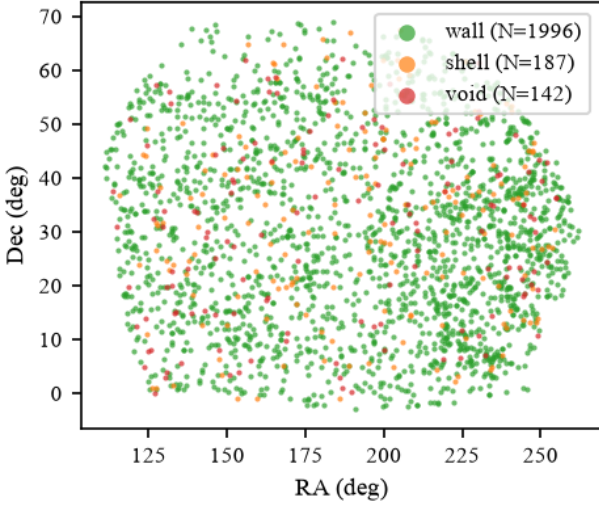


Figure 2. Masked RA–Dec for cosmology-grade SNe: wall ($N = 1996$, green), shell ($N = 187$, orange), void ($N = 142$, red). Outside-mask and edge-excluded objects omitted.

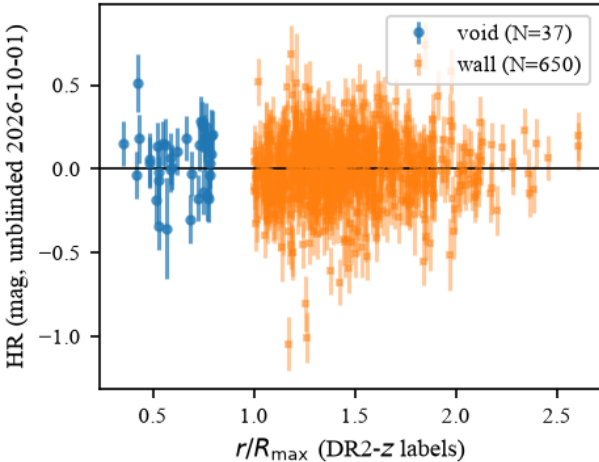


Figure 3. Hubble residuals vs $r/R_{\max}$ (unblinded 2026-10-01) for the full-volume DR2 sample: void (red circles, $N = 37$, $n = 33$ matched) and wall (green squares, $N = 650$) with per-SN errors (SALT2+0.12 intrinsic). No trend with environment is visible.

~ 300 km/s bounds the coherent term at $\lesssim 0.02$ mag; $z_{\min}$ grid spread < 0.002 mag; diagonal 0.055 mag retained. Systematics budget (Table 3):

6 DISCUSSION AND CONCLUSIONS

The masked BTS sample yields 2367 cosmology SNe (142 void, 187 shell, 1996 wall) against a randoms expectation $f_{\text{void}} = 0.190$: an apparent void/wall ratio 0.24 overall (0.26 Ia, 0.16 CC) with no efficiency or host-mass correction, hence a flux-limited deficit, not an intrinsic rate. The in-mask transient fraction $2465/6873 = 0.359$ exceeds the geometric sky fraction 0.177 because BTS pointings and depth overlap the SDSS northern footprint non-uniformly; randoms therefore predict volumes, not detections; completeness reweighting shifts ratios only +3–9%, bounding the magnitude-limit effect short of a full efficiency function. The void redshift offset (median 0.064 vs 0.056) is the Malmquist gradient in the same data. Primary Hubble test is full-volume DR2- z (752 in-mask, 37 void): matched $\Delta_{\text{HR}} = +0.034 \pm 0.030$ mag (stat, $n = 33$), ± 0.042 with systematics, consistent with zero (two-sided 95% interval $[-0.05, +0.12]$ mag). The BTS-gated branch agrees ($+0.023 \pm 0.034$, $n = 27$); joint fit recovers the mass slope (-0.065 ± 0.011) — a null with an understood confound. Limitations: 1-deg mask quantization, TNS-key join unavailable (HR arm purged via BTS+DR2 strings, 0/396 peculiar; counts arm carries BTS-string audit only), PV flow-model bounded analytically, and an FRB arm that is a forecast only (Catalog 2 required). Unblinding protocol: thresholds passed full-volume nulls (RA-shuffle $p = 0.27$, bootstrap 0.028 vs 0.030, injections exact), a closed systematics budget, and frozen cuts; key rotated at unblinding; unblinded 2026-10-01: matched offset $+0.034 \pm 0.030$ mag (stat), ± 0.042 with systematics, consistent with zero; two-sided 95% interval $[-0.05, +0.12]$ mag at this sample size.

FRB stacking outlook

FRB error budget (Cat1, following Macquart et al. (2020) and void-decrement simulations (Walker et al. 2024)): per-sightline $\sigma_{\text{tot}} \sim 150\text{--}200$ pc cm $^{-3}$ (IGM 80–120, host 80–150, Milky Way residual 20–40 with NE2001–YMW16 median -3.3 , std 37.5, p90 68.1, halo 20). Void signal $\sim 10\text{--}30$. A $\delta\text{DM} < 50$ limit at 95 per cent needs $N_{\text{void}} \gtrsim 36$ (~ 200 localised total); unlocalised $z < 0.114$ needs ~ 900 void sightlines, have $\sim 30\text{--}50$.

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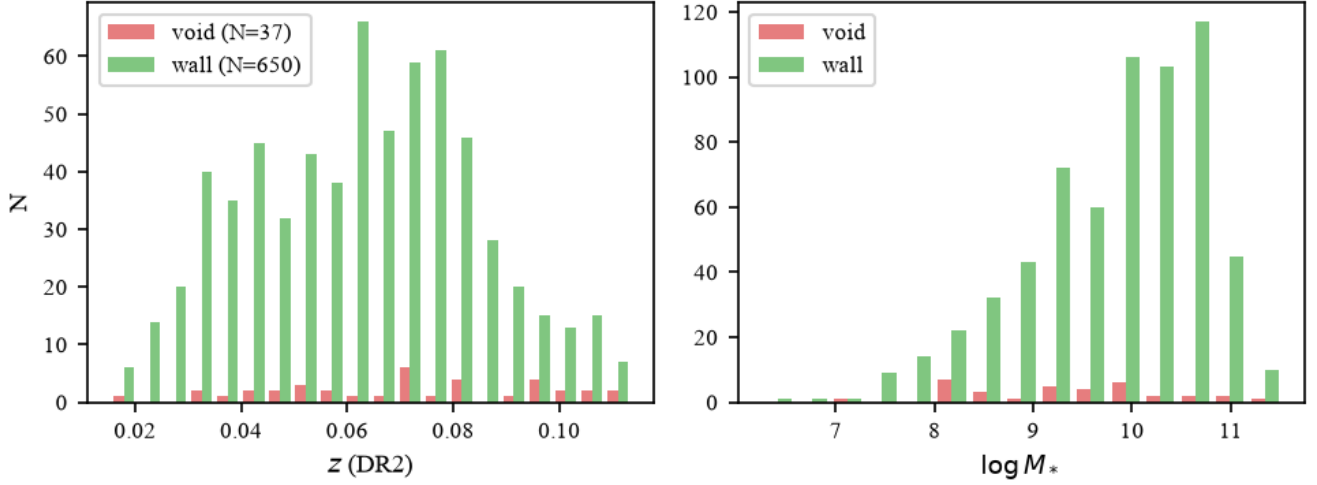


Figure 4. Redshift (left) and host stellar mass (right) distributions for void (red, $N = 37$) vs wall (green, $N = 650$) in the full-volume DR2 sample (47 shell + 15 outside + 3 unclassified omitted from plot). Void hosts are lower-mass (means 9.37 vs 9.81) and slightly higher- z , motivating mass+ z matching.

Table 3. Δ_{HR} systematics budget (full-volume DR2- z primary analysis). Stat error 0.030 mag dominates; systematics summed below are bounds, not detections. Sources: logs/dr2_nulls.log, logs/dr2_pv.log, logs/step_zcut.log, logs/mask_buffer_h.log.

Source	Bound (mag)	Type	Status
RA-shuffle null (fullx)	mean +0.002, $p = 0.27$	test	pass
Block bootstrap vs analytic (fullx)	0.028 vs 0.030	test	agrees
Injection 0.00/0.05 (fullx)	recovers exactly	test	pass
Mass linear vs step	$[0.026 - 0.032] = 0.006$	bound	bounded
z_{min} grid spread	< 0.004	bound	bounded
CMB-frame shift	0.005	bound	bounded
Mask edge erosion	2.5% on fraction	bound	subdom.
Spec- z -only cross-check (fullx)	$+0.041 \pm 0.042$ vs $+0.034 \pm 0.030$	test	agrees
Full-volume DR2- z cross-check	$+0.034 \pm 0.030$ vs $+0.023 \pm 0.034$	test	agrees
BTS- z label blur	void precision 10/31 in DR2 overlap	bound	bounded
Subtype purge (HR sample, BTS+DR2)	0/396 peculiar	test	pass
PV flow model	corrected, Δ unchanged	bound	bounded
BTS efficiency/BBC	~ 0.01 mag bound	bound	bounded
BTS completeness reweight	shifts ratios +3–9%	test	agrees
Combined systematic	~ 0.029 (quadrature)	total	–
Total (stat+syst)	~ 0.042	total	–

Spark 1.3) assisted in building analysis codes, LaTeX files, and figures; however, the scientific analysis, interpretations, and conclusions remain the sole responsibility of the author, who independently verified all generated numbers by re-execution.

<https://github.com/hello2himel/VOIDS/releases/tag/v0.44-primary-branch>; repository public at
<https://github.com/hello2himel/VOIDS>; Zenodo DOI
to follow via GitHub integration.

7 DATA AVAILABILITY

VAST <https://zenodo.org/records/7406035> (DOI <https://doi.org/10.5281/zenodo.7406035>, v1.3.0, zip sha256 9896002c...acef9); BTS <https://sites.astro.caltech.edu/ztf/bts/explorer.php> query quality+y purity+y (2026-09-29 UTC, CSV sha256 77668fe8...f38d65); CHIME Catalog 1 Vizier <https://vizier.cds.unistra.fr/viz-bin/VizieR-3?-source=J/ApJS/257/59/table2>; derived b3_match_v2.parquet (sha256 1a162c74...af77530) with scripts and randoms (seed 42/43/44) archived at

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